

NADMO Integrated Warehouse & Logistics Management System (NADMO-WMS)

Product Requirements Document — Version 2.0

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Table of Contents

- 1. Strategic Context & Problem Statement
- 2. Product Vision
- 3. Product Form Factor
- 4. Target Users & Personas
- 5. Product Value Proposition
- 6. Scope Boundaries
- 7. Real-World Use Cases
- 8. Core User Flow
- 9. System Architecture
- 10. Feature Inventory
- 11. Key Page Layouts
- 12. Detailed Module Specifications
- 13. Data Model
- 14. API Design
- 15. Security Architecture
- 16. Compliance & Governance
- 17. Design System & User Experience
- 18. Connectivity Resilience
- 19. Ghana National Map & GIS
- 20. Copy & Language Standards
- 21. Non-Functional Requirements
- 22. Implementation Roadmap
- 23. Success Metrics & KPIs
- 24. Risk Register
- 25. Quality Assurance & Testing Strategy
- 26. Training & Change Management
- 27. Support & Maintenance

28. Budget & Estimation Framework

29. SMS Notification Templates

30. Bulk Data Import Specifications

31. Open Items & Decisions Log

32. Appendices

33. Conclusion

Executive Summary

The **NADMO Integrated Warehouse & Logistics Management System (NADMO-WMS)** is a sovereign, secure, and intelligent enterprise platform designed to transform Ghana’s national disaster preparedness and response capability. It will connect every NADMO warehouse — from the national strategic stockpile in Greater Accra through all 16 regional hubs to all 261 district warehouses — into a single, real-time logistics nervous system.

Unlike fragmented spreadsheet tracking, paper registers, or phone-based coordination, NADMO-WMS provides:

- **Total Visibility:** Real-time stock levels, movements, and status across every warehouse, accessible from headquarters to the district level.
- **Intelligent Coordination:** AI-assisted demand forecasting, automated stock redistribution recommendations, and smart approval workflows.
- **Immovable Accountability:** An immutable, append-only audit trail for every item movement, user action, and decision.
- **Field-Ready Operations:** Progressive Web App (PWA) and future native mobile app with offline sync, barcode/QR scanning, GPS tracking, and digital signatures.
- **Government-Grade Security:** End-to-end encryption, role-based access control (RBAC), multi-factor authentication (MFA), and full compliance with Ghana’s Data Protection Act and NITA standards.

This document defines the product vision, functional and non-functional requirements, system architecture, data model, security framework, compliance posture, implementation roadmap, and success metrics for delivering an award-winning, procurement-ready government logistics platform.

1. Strategic Context & Problem Statement

1.1 National Context

Ghana faces a diverse and intensifying disaster risk profile, including:

- **Flooding** — annual seasonal flooding in northern, central, and coastal regions
- **Droughts and food insecurity** — particularly in northern savannah zones
- **Bush and industrial fires** — across urban and rural districts
- **Earthquakes and seismic risks** — especially in Accra and the southeast
- **Epidemics and public health emergencies** — cholera, meningitis, COVID-19-type events
- **Storm surges and coastal erosion** — affecting fishing communities

NADMO is the statutory lead agency for disaster prevention, preparedness, response, and recovery. Effective response depends on the right relief materials being in the right place at the right time — a challenge that current manual processes cannot reliably meet.

1.2 Current Pain Points

Pain Point	Operational Impact	Risk to Lives
Paper-based stock registers	Delayed reporting, transcription errors, lost records	Supplies may not arrive where needed
No real-time national visibility	HQ cannot see regional/district stock during emergencies	Slow, reactive response

Manual phone/WhatsApp coordination	Miscommunication, no audit trail, duplication	Resources misallocated
No predictive stock planning	Stockouts before disasters; waste from overstocking	Communities left without aid
Weak audit trail	Pilferage, disputes, donor reporting gaps	Loss of public and donor trust
Limited field mobility	Field officers cannot update systems in real time	Delays in confirming receipts/dispatch

1.3 Strategic Objectives

#	Objective	Measurable Outcome
1	Establish a single source of truth for all NADMO warehouse stock	100% of warehouses digitised within 12 months
2	Enable real-time visibility of stock and movements nationwide	Dashboard refreshed within 5 minutes of any transaction
3	Reduce response time to disaster-affected communities	50% reduction in dispatch-to-delivery time within 18 months
4	Strengthen accountability and reduce losses	100% traceability of all items; zero unrecorded movements
5	Improve donor and government reporting	Donor-ready reports generated in under 2 minutes
6	Build institutional resilience and capacity	90%+ user adoption across all warehouses within 6 months of rollout

2. Product Vision

To make NADMO the most transparent, responsive, and technologically advanced disaster logistics organisation in West Africa — where every relief item is tracked, every movement is visible, and no community is left waiting because of information failure.

3. Product Form Factor

3.1 Primary Delivery: Progressive Web Application (PWA)

Item	Detail
Selected Form	Progressive Web App (PWA)
Rationale	Single codebase; works on desktop, tablet, and mobile; offline-capable; no app store dependency; government IT can self-host
First-Class Devices	Desktop at HQ/Regional offices; Android tablets at district warehouses; smartphones for field supervisors
Phase 2	Native Android/iOS apps with full offline synchronisation and hardware barcode scanning
Phase 3	Deep integration with Ghana Government ERP (GIFMIS), NEMA, Ghana Meteorological Agency, Ghana Health Service, and donor systems

3.2 Why PWA for Government

- **Sovereignty:** No dependency on Apple/Google app stores for deployment or updates.
- **Reach:** Works on the lowest common denominator devices already in use.
- **Resilience:** Service workers enable offline data capture and background sync.
- **Maintainability:** One codebase reduces total cost of ownership (TCO).
- **Security:** Served over HTTPS with controlled deployment pipelines.

4. Target Users & Personas

Role	Profile	Core Pain Today	Key Needs
Director-General / HQ Executive	Senior NADMO official; strategic decision-maker; presents to Cabinet and donors	No real-time national visibility; relies on phone calls and weekly reports	National dashboard, one-click reports, critical alerts, drill-down to district
HQ Logistics & Procurement Officer	Coordinates national stock, procurement, and large transfers	Cannot track in-transit shipments; no automated low-stock alerts	Transfer order management, vehicle tracking, procurement triggers, stock forecasting
Regional Warehouse Manager	Manages regional hub + supervises district warehouses in region	No structured way to receive/dispatch; paper records	Regional dashboard, district oversight, approval workflows, discrepancy resolution
District Warehouse Officer	Frontline officer at district capital warehouse	No digital tools; notebook records; delays during crises	Mobile-friendly stock intake/dispatch, QR scanning, receipt confirmation, offline sync
Field Response Officer	Deploys to disaster sites with relief materials	Cannot update status in the field	Mobile dispatch confirmation, GPS tracking, photo evidence, digital signatures
Auditor / Finance Officer	Reviews inventory records and asset utilisation	Manual reconciliation takes weeks; weak audit trail	Immutable audit logs, exception reports, valuation reports, donor reconciliation
System Administrator (NITA/NADMO IT)	Manages users, roles, security	Needs enterprise-grade controls	RBAC, MFA, audit logs, backups, API management

5. Product Value Proposition

5.1 For Users

- **Real-time visibility** of every item across every warehouse, from Paga to Axim.
- **Automated alerts** when stock falls below critical thresholds — before disasters strike.
- **Digital audit trail** — no lost paperwork, no disputed records, no unaccounted movements.
- **Field-ready interface** that works on basic Android tablets and smartphones, even with poor connectivity.
- **Intelligent recommendations** for stock redistribution based on disaster risk, seasonality, and consumption patterns.

5.2 For Government of Ghana

- Positions NADMO as a modern, accountable, data-driven institution.

- Reduces waste, pilferage, and expiry losses through full traceability and FEFO (First-Expiry-First-Out) logic.
- Supports national disaster preparedness planning with evidence-based pre-positioning.
- Strengthens inter-agency coordination with standardised data sharing.

5.3 For Development Partners & Donors

- Instant, donor-ready reports in World Bank, UN OCHA, USAID, and EU formats.
- Full transparency on utilisation of donated goods.
- Verifiable impact metrics for disaster response operations.

6. Scope Boundaries: What We Are NOT Building

Out of Scope	Reason	Future Integration Point
Financial payments / procurement invoicing	Handled by GIFMIS and NADMO finance	API integration in Phase 3
HR management and payroll	Separate NADMO HR systems	SSO integration possible
Disaster incident command & field operations	Separate CAD/incident management system	Incident-linking module in Phase 2
Public-facing citizen portal	Not part of v1; requires policy approval	Phase 3 citizen feedback module
Full IoT sensor network	Cold-chain monitoring; phased due to hardware cost	Phase 2 for strategic warehouses
National ID integration	Outside logistics scope; NIA integration optional	Phase 3 for beneficiary verification

7. Real-World Use Cases

Use Case 1: Flood Early Response (District Officer)

A district officer in Accra Metro receives a dispatch order from HQ for 200 emergency food packs. She logs into NADMO-WMS on her tablet, scans the QR codes on the arriving pallets, confirms receipt, updates stock levels, and captures a digital signature. The system automatically notifies HQ and the regional manager. The entire transaction is timestamped and geo-tagged.

Use Case 2: National Stock Audit (Director-General)

The Director-General opens NADMO-WMS before a Cabinet briefing. A single national dashboard shows total stock across all warehouses, items in transit, warehouses below critical threshold (flagged red), and the last 30 days of movement history — all without calling a single officer. He exports a donor-ready PDF in two clicks.

Use Case 3: Cross-Region Emergency Transfer (HQ Logistics)

After a major flooding event in Bono Region, HQ logistics redistributes 1,000 emergency tents from the Greater Accra central warehouse to Sunyani. The officer creates a transfer order, the system routes it for approval based on value/quantity rules, assigns a vehicle and driver, and generates a digital waybill with QR code. The receiving warehouse scans the waybill on arrival. The in-transit shipment is tracked on a live map until confirmed delivery.

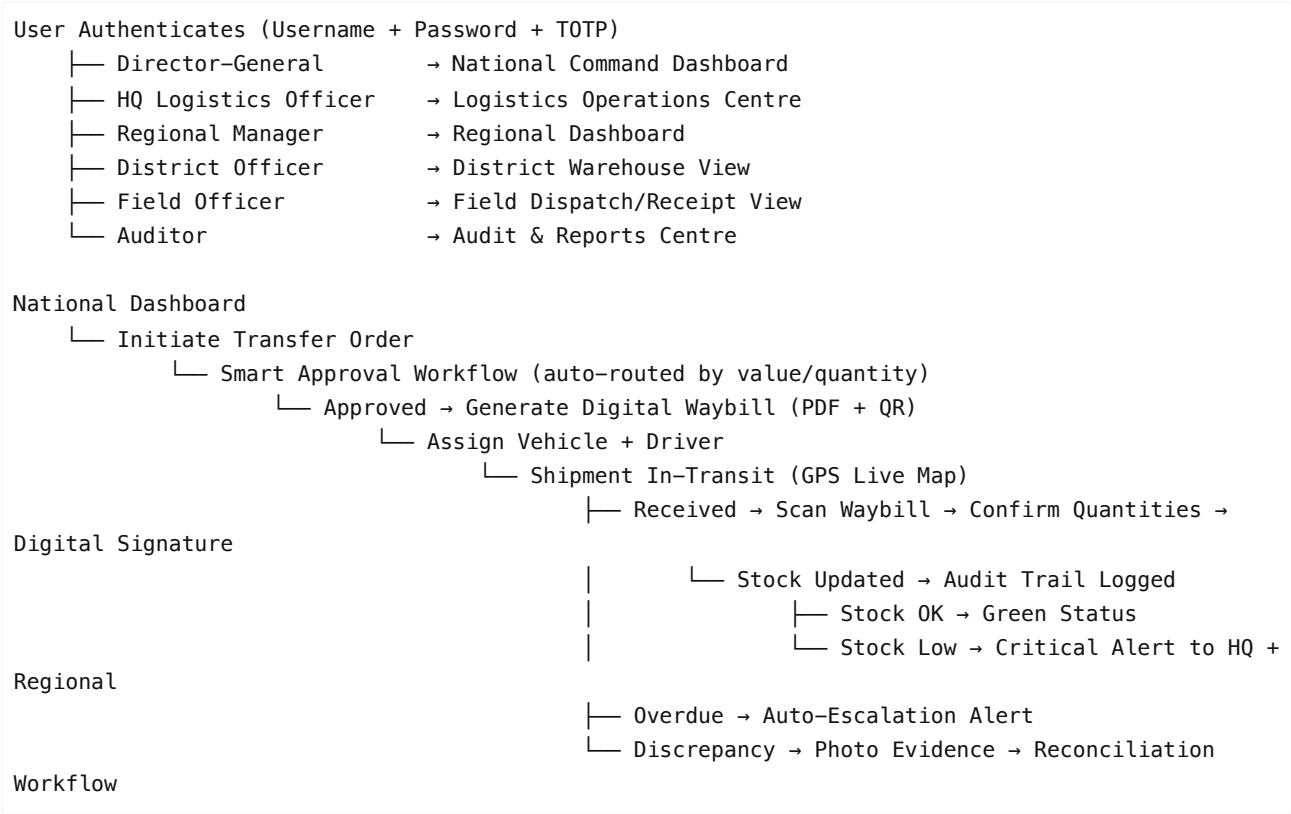
Use Case 4: Predictive Pre-Positioning (HQ Analyst)

Before the rainy season, the system’s analytics engine recommends pre-positioning additional water purification tablets and tarpaulins in the Upper East, Savannah, and Northern regions based on historical flood patterns, current stock levels, and seasonal risk indices. HQ approves the recommendation and dispatches materials accordingly.

Use Case 5: Discrepancy Resolution (Regional Manager)

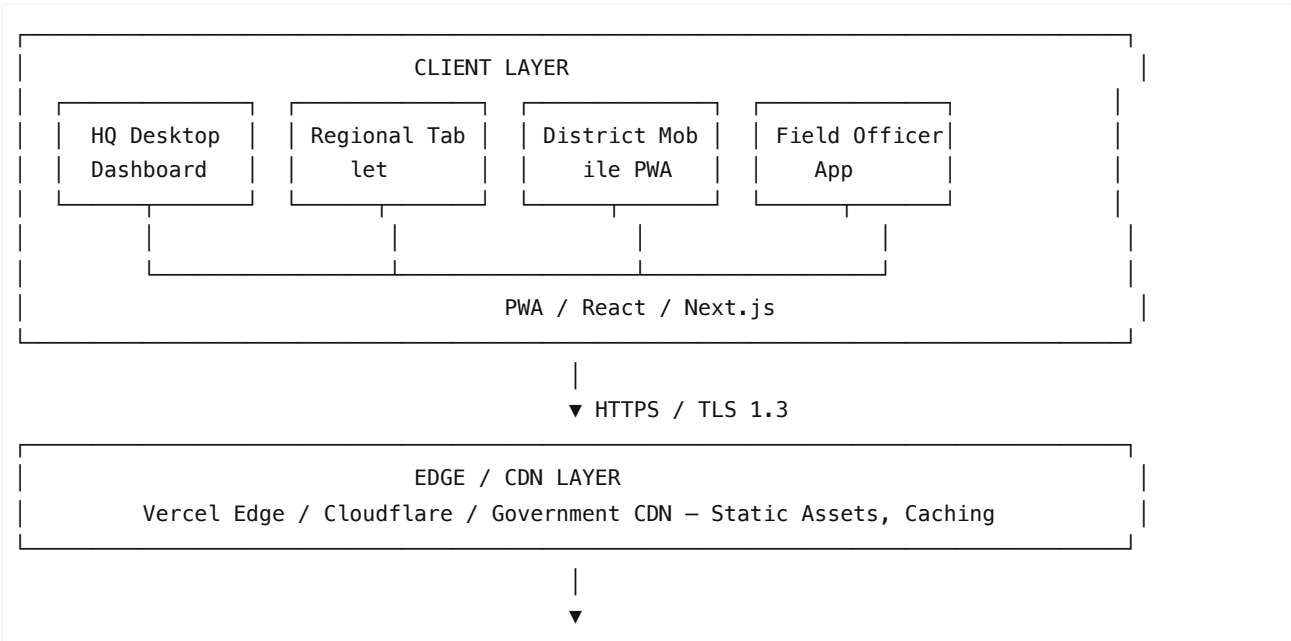
A district warehouse reports receiving 180 blankets instead of 200 dispatched. The system automatically flags a discrepancy, notifies the regional manager and HQ logistics, and triggers a digital reconciliation workflow. Photos, driver statements, and witness signatures are attached before the case is closed.

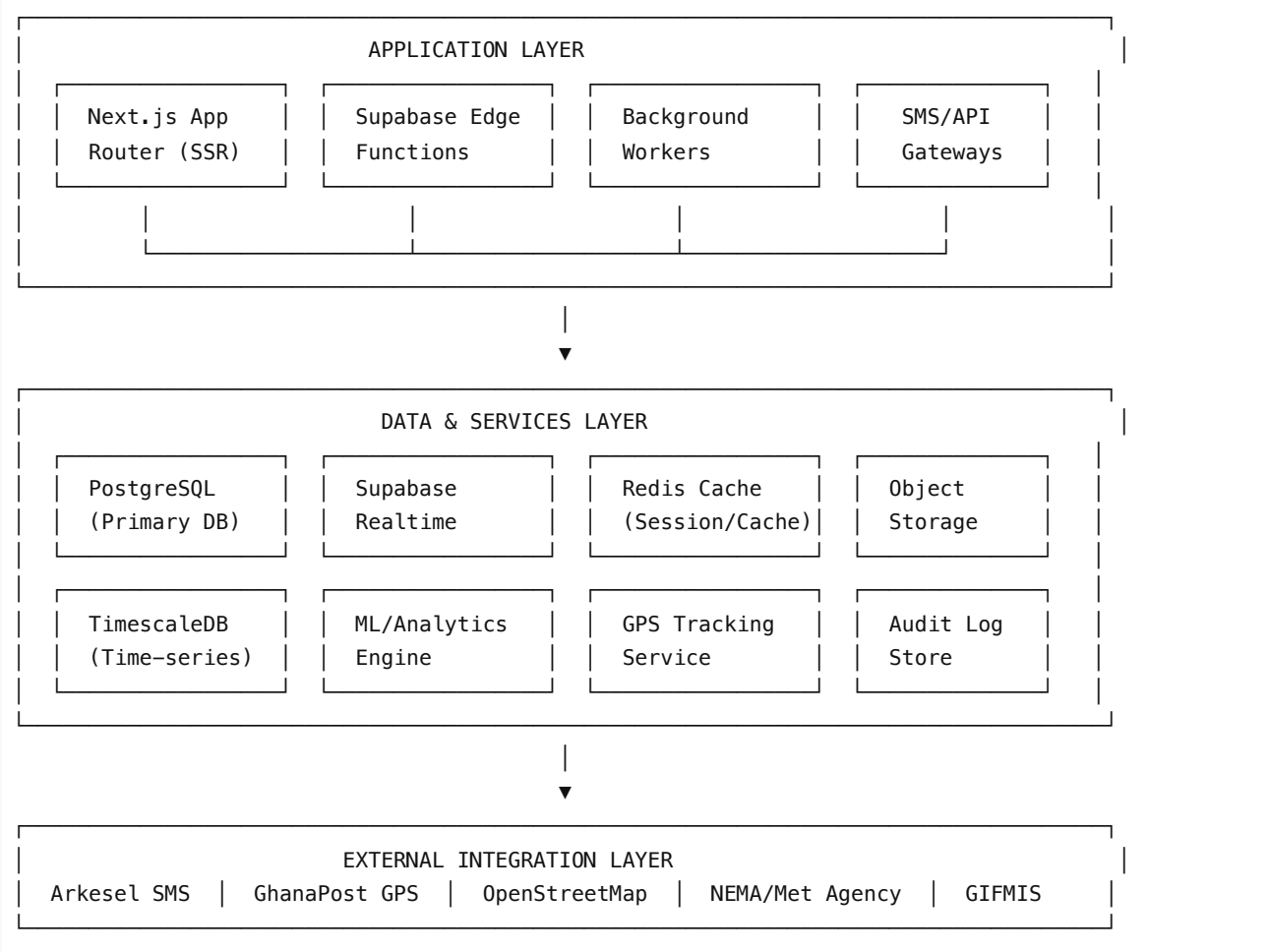
8. Core User Flow



9. System Architecture

9.1 High-Level Architecture





9.2 Technology Stack

Layer	Technology	Justification
Frontend	Next.js 14+ (App Router), TypeScript, Tailwind CSS	Server-side rendering, PWA support, excellent developer ecosystem
State Management	Zustand + TanStack Query	Lightweight, optimised for server-state and caching
UI Components	shadcn/ui + Radix UI	Accessible, customisable, government-grade design system
Backend	Supabase (PostgreSQL + PostgREST + GoTrue)	Open-source, self-hostable, built-in RLS, real-time subscriptions
Edge Functions	Supabase Edge Functions (Deno)	Low-latency serverless functions for SMS, PDF generation, integrations
Database	PostgreSQL 15+ with Row-Level Security	ACID compliance, rich JSON support, geospatial extensions (PostGIS)
Time-Series	TimescaleDB extension	Optimised for tracking stock levels, movements, and sensor data over time
Caching	Redis / Upstash	Session storage, dashboard cache, rate limiting
Object Storage	Supabase Storage / MinIO	Waybill PDFs, signatures, photos, import files

Maps	Mapbox GL JS / Leaflet + OpenStreetMap	Ghana boundary data, offline-capable, no licensing lock-in
GPS Tracking	Custom GPS service + GhanaPost GPS integration	Vehicle tracking, geofencing, route optimisation
SMS	Arkesel API	Ghana-local provider, branded sender ID, reliable delivery
Email	Resend / AWS SES / Government SMTP	Transactional and scheduled reports
Monitoring	Sentry + Logflare + UptimeRobot	Error tracking, performance monitoring, uptime alerts
CI/CD	GitHub Actions	Automated testing, security scanning, deployment

9.3 Deployment Strategy

Phase	Environment	Hosting	Notes
Phase 1 (Months 1–6)	Production + Staging	Vercel + Supabase Cloud	Rapid deployment, automatic scaling, global CDN
Phase 2 (Months 7–12)	GovTech Ghana Migration Readiness	Containerised deployment (Docker/Kubernetes)	Portable, air-gappable, sovereign data option
Phase 3 (Year 2+)	Sovereign / Hybrid Cloud	Government Data Centre or accredited local cloud	Full data residency, NITA compliance

9.4 Portability & Sovereignty

The architecture is intentionally cloud-agnostic:

- No vendor-proprietary runtime features.
- All infrastructure endpoints injected via environment variables.
- Database schema is standard PostgreSQL with no Supabase-specific lock-in.
- Supabase stack is fully open-source and self-hostable.
- Migration path: DNS cutover → container deployment → data export/import → SSL provisioning → security audit → go-live.

10. Feature Inventory

10.1 CORE (MVP — Must Have for Launch)

NADMO-WMS
└─ Module 1: Authentication, Identity & Access Management
├─ Secure login (username + password + TOTP 2FA)
├─ Role-based access control (RBAC) with least privilege
├─ Single Sign-On (SSO) readiness (SAML 2.0 / OIDC)
├─ Session timeout + concurrent session control
├─ Password policy enforcement (min 12 chars, complexity, 90-day expiry)
├─ Activity logging for all authentication events
└─ Account deactivation / suspension workflows
└─ Module 2: National Command Dashboard (HQ)
├─ Real-time national stock map (Ghana choropleth – regions + districts)

- KPI tiles: Total SKUs, In-Transit shipments, Critical-stock warehouses, Active transfers
- Critical stock alerts panel (red/amber/green threshold indicators)
- Recent activity feed (last 100 transactions nationwide)
- Quick-action buttons: Initiate Transfer, Raise Alert, Export Report
- Drill-down: National → Regional → District → Warehouse
- Executive briefings generator (one-click PDF summary)

— Module 3: Regional & District Warehouse Dashboards

- Regional view of all district warehouses in region
- District warehouse operational view
- Stock-on-hand by SKU category
- Incoming/outgoing transfer queues
- Local alert panel and notification centre
- Performance scorecard (receipt timeliness, accuracy)

— Module 4: Warehouse Inventory Management

- Stock intake (receive from HQ / external supplier / donor)
- Stock dispatch (outbound to field / another warehouse / disaster site)
- Real-time stock level per SKU per warehouse
- Stock adjustment (write-off, damage, expiry, correction with mandatory reason)
- Batch/lot tracking and expiry date management (FEFO)
- Barcode / QR code generation and scanning (PWA camera + optional Bluetooth scanner)
- Minimum stock threshold configuration per warehouse per SKU
- Cycle count and physical stock reconciliation

— Module 5: Transfer Order & Waybill Management

- Create transfer order (source → destination warehouse)
- Hierarchical approval workflow (District → Regional → HQ → Director)
- Digital waybill generation (PDF export with QR code and barcode)
- Vehicle and driver assignment
- In-transit status tracking with GPS live map
- Receipt confirmation + digital signature + photo capture
- Discrepancy reporting and reconciliation workflow
- Delivery SLA monitoring with auto-escalation

— Module 6: Alerts & Notifications

- Critical stock level alerts (configurable thresholds)
- Overdue shipment alerts (configurable SLA hours)
- Discrepancy alerts (escalated to regional + HQ)
- Approval SLA breach alerts
- In-app notification centre
- SMS notifications via Arkesel
- Email notifications for reports and escalations
- Push notifications (PWA + future native app)

— Module 7: Hierarchical Supervision & Audit Trail

- Full immutable transaction log (who, what, when, from where, IP)
- HQ supervisory view – drill down: National → Regional → District
- Stock movement history per SKU per warehouse
- User activity log (login, edits, exports, approvals)
- Audit log exports (PDF/CSV) for auditors
- Tamper-evident log hashing

— Module 8: Master Data Management

- SKU catalogue (manual entry + bulk import)
- Item categorisation (Food / Shelter / Medical / Equipment / PPE / Water & Sanitation)

- └─ Unit of measure configuration
- └─ Item photos, descriptions, and hazard classifications
- └─ Warehouse directory (HQ, 16 regional, 261 district warehouses)
- └─ Warehouse profile (address, capacity, manager, contact, GPS coordinates)
- └─ Regional hierarchy mapping
- └─ User directory and role assignment

10.2 IMPORTANT (Post-Launch Iterations — Months 4–9)

- └─ Module 9: Reporting & Business Intelligence
 - └─ Predefined reports: Stock Status, Movement History, Low Stock, Transfer Summary
 - └─ Scheduled report delivery (daily/weekly/monthly email)
 - └─ Custom report builder (filter by region, district, SKU, date range)
 - └─ Export to PDF, Excel, CSV
 - └─ Donor-ready report templates (UN OCHA, World Bank, USAID, EU)
 - └─ Executive KPI dashboard
- └─ Module 10: Advanced Analytics & Predictive Intelligence
 - └─ Demand forecasting by region/season/disaster type
 - └─ Stock redistribution recommendations
 - └─ Expiry risk analysis and alerts
 - └─ Consumption trend analysis
 - └─ Seasonal risk correlation (rainy season, dry season, Harmattan fires)
 - └─ What-if scenario modelling
- └─ Module 11: Warehouse & Location Management
 - └─ Warehouse capacity and utilisation tracking
 - └─ Warehouse status flags (Operational / Limited / Closed)
 - └─ Storage zone/bin management
 - └─ Warehouse condition assessments
 - └─ Regional hierarchy mapping with boundary visualisation
- └─ Module 12: User & Organisation Management
 - └─ User directory (create, edit, deactivate accounts)
 - └─ Role assignment and permission matrix
 - └─ Warehouse-to-user mapping
 - └─ Bulk user import (CSV/Excel)
 - └─ Delegation rules (leave/backup approvers)
 - └─ Training mode / sandbox environment
- └─ Module 13: Offline-First Mobile Operations
 - └─ Native Android app with offline sync
 - └─ Background data synchronisation
 - └─ Conflict resolution for concurrent edits
 - └─ Hardware barcode/QR scanner support
 - └─ GPS tracking and geofencing
- └─ Module 14: Quality & Compliance
 - └─ Cold-chain monitoring dashboard (Phase 2 IoT)
 - └─ Donor-specific reporting requirements
 - └─ Beneficiary distribution tracking
 - └─ Compliance checklists and certifications
 - └─ Integration with Ghana Health Service for medical stock

10.3 FUTURE ROADMAP (Year 2+)

- Native iOS app and enhanced offline capabilities
- Full IoT integration (temperature, humidity, door sensors)
- Advanced GPS vehicle tracking and route optimisation
- Disaster incident linking (shipments tied to active incidents)
- Open API for Ghana Government ERP integration (GIFMIS)
- Beneficiary registration and aid distribution verification
- Integration with NEMA, Ghana Meteorological Agency, Ghana Hydrological Authority
- Satellite imagery / flood monitoring integration
- Multi-language support (Twi, Ga, Ewe, Hausa, Dagbani)
- Blockchain-anchored audit trail (optional for donor transparency)

11. Key Page Layouts

11.1 National Command Dashboard (HQ)

[NADMO Logo] NADMO WAREHOUSE COMMAND CENTRE						[🌐 EN] [🔔 3] [👤 DG ▼] [⚙️] [📱]	
DASHBOARD INVENTORY TRANSFERS ALERTS REPORTS ANALYTICS WAREHOUSES USERS SETTINGS [+ NEW TRANSFER]	NATIONAL OPERATIONS OVERVIEW					[Today ▼] [Export]	
	261	14	23	8	98.2%		
	Warehouses	In-Transit	Critical	Active	Stock		
	Active	Shipments	Stock	Transfers	Accuracy		
	GHANA NATIONAL MAP (interactive choropleth map – green/amber/red by stock health) [Zoom] [Layer: Region]			🚨 CRITICAL ALERTS Brong-Ahafo: Tents < 5% Volta: Food Packs 0 Oti: Medical Kits 10% [View All Alerts →]			
				📊 PREDICTIVE INSIGHTS ↑ 23% flood risk in Northern Region Recommend pre-positioning: • Water tablets: +5,000 units • Tarpaulins: +1,200 units			
	📈 STOCK TRENDS (30 DAYS) [Line chart]			🚚 RECENT ACTIVITY 10:42 Accra → Sunyani (In Transit) 10:31 Receipt confirmed: Tamale 09:55 Low stock alert: Kumasi 09:12 Waybill generated: Wa [View Full Log →]			

11.2 District Warehouse View

[NADMO Logo] DISTRICT WAREHOUSE: ACCRA METRO [🔄 Sync] [🔔] [👤 ▼]	
	STOCK ON HAND TODAY

RECEIVE
DISPATCH
TRANSFERS
ALERTS
REPORTS

[+ QUICK
ACTION]

Search items... [Category ▼] [Scan QR 📄]

Item	Category	Quantity	Expiry / Status
Food Packs	Food	1,240	✅ OK
Tents	Shelter	42	❌ CRITICAL
Medical Kits	Medical	89	⚠️ LOW
Water Tabs	WASH	5,600	✅ OK
Blankets	Shelter	310	✅ OK

INCOMING TRANSFERS
TRF-2026-0041 (2)
[Confirm Receipt]

OUTGOING TRANSFERS
TRF-2026-0038 (In Transit)
[Track Vehicle]

12. Detailed Module Specifications

12.1 Module 1: Authentication, Identity & Access Management

12.1.1 Authentication Flow

- User navigates to `https://wms.nadmo.gov.gh`.
- Enters username and password.
- System validates credentials against Supabase Auth.
- If MFA is enabled (required for HQ/Regional/Director roles), prompt for TOTP code from authenticator app.
- On success, system issues JWT access token (short-lived, 15 minutes) and refresh token (secure, httpOnly cookie).
- User redirected to role-appropriate dashboard.
- All login events logged with IP address, user agent, timestamp, and MFA status.

12.1.2 Multi-Factor Authentication (MFA)

Role	MFA Required	Method
Director-General / Deputy	Yes	TOTP (Google Authenticator, Microsoft Authenticator, Authy)
HQ Logistics / Procurement	Yes	TOTP
Regional Managers	Yes	TOTP
District Officers	Recommended	TOTP or SMS OTP
Auditors	Yes	TOTP
System Administrators	Yes	TOTP + hardware key optional

12.1.3 Role-Based Access Control (RBAC)

Role	Code	Description
System Administrator	sysadmin	Full platform administration, user management, security settings
Director-General	dg	National oversight, executive reports, major transfer approval
HQ Logistics Officer	hq_logistics	National transfers, procurement coordination, threshold config

HQ Procurement Officer	hq_procurement	Supplier intake, donor receipt, stock valuation
Regional Manager	regional_manager	Regional oversight, medium transfer approval, district supervision
District Warehouse Officer	district_officer	District stock intake/dispatch, receipt confirmation
Field Response Officer	field_officer	Field dispatch, mobile updates, photo capture
Auditor	auditor	Read-only access to all data, audit log exports
Read-Only Observer	readonly	View-only access scoped to assigned warehouses

12.1.4 Session Security

Control	Implementation
Access token lifetime	15 minutes
Refresh token lifetime	7 days (with rotation)
Inactivity timeout	30 minutes (configurable per role)
Concurrent sessions	Maximum 2 active sessions per user; new login invalidates oldest
Password policy	Min 12 characters; uppercase, lowercase, number, special character; 90-day expiry; no reuse of last 6 passwords
Account lockout	5 failed attempts → 15-minute lockout; alert security admin
Password reset	Email/SMS OTP with 1-hour expiry; admin-initiated reset logged

12.2 Module 4: Warehouse Inventory Management

12.2.1 Stock Intake Workflow

1. User selects **Receive Stock**.
2. System prompts for source: **HQ Transfer / External Supplier / Donor / Other Warehouse / Return from Field**.
3. User scans QR/barcode on incoming items or selects SKU manually.
4. User enters quantity, batch/lot number, manufacturing date, expiry date (if applicable), storage location.
5. System validates against expected inbound transfer (if linked) or creates new intake record.
6. User uploads photo evidence (optional but recommended for donors/large receipts).
7. User confirms; stock updated in real time; audit log entry created.
8. If linked to transfer order, transfer status updated to **Received**.

12.2.2 Stock Dispatch Workflow

1. User selects **Dispatch Stock**.
2. System prompts for destination: **Another Warehouse / Field Site / Disposal / Donor Return**.
3. User scans items or selects SKU; system checks available stock (respecting FEFO for expiry-managed items).
4. User enters quantity and reason.
5. System validates sufficient stock; warns if dispatch would breach minimum threshold.
6. For transfers to another warehouse, a transfer order is auto-created or linked.
7. Stock is **allocated** (reserved) at dispatch and **deducted** on waybill confirmation.
8. Audit log entry created.

12.2.3 Stock Adjustment Workflow

Adjustment Type	Approval Required	Reason Mandatory	Examples
Damage	District Officer + Regional Manager	Yes	Broken water containers, torn tents
Expiry	District Officer	Yes	Expired medical kits, food packs
Correction	District Officer + Regional Manager	Yes	Data entry error, cycle count variance
Write-off	Regional Manager + HQ Logistics	Yes	Obsolete equipment, disposal
Loss / Theft	Regional Manager + HQ Logistics + Auditor	Yes	Missing items, investigation required

All adjustments require photo evidence for quantities > 20 units or value > GHS 1,000.

12.2.4 Batch, Lot & Expiry Tracking (FEFO)

- Every stockable item can optionally have a batch/lot number and expiry date.
- System enforces **First-Expiry-First-Out (FEFO)** for dispatches.
- Expiry alerts generated 90, 60, 30, and 7 days before expiry.
- Expired items are automatically quarantined from dispatch and flagged for disposal.

12.3 Module 5: Transfer Order & Waybill Management

12.3.1 Transfer Order Lifecycle

Status	Trigger	UI Indicator	Who Can Act
Draft	Created but not submitted	Grey	Creator only
Pending Approval	Submitted, awaiting sign-off	Amber pulsing	Assigned approver
Approved	Approved by authorised officer	Blue	Logistics / source warehouse
Ready for Dispatch	Vehicle and driver assigned	Blue	Source warehouse
In Transit	Vehicle departed	Blue + animated route	Read-only; driver/field officer updates
Received	Destination confirmed receipt	Green	Destination officer
Discrepancy	Quantity/quality mismatch on receipt	Red	Regional Manager + HQ Logistics
Cancelled	Cancelled before delivery	Grey strikethrough	HQ Admin or creator (if not yet dispatched)
Overdue	In-transit > configured SLA	Red flashing	Auto-escalated

12.3.2 Hierarchical Approval Matrix

Transfer Scale	Quantity	Est. Value	Approver	SLA
Routine	< 100 units	< GHS 5,000	District Officer (self-authorising)	Immediate
Standard	100–499 units	GHS 5,000–49,999	Regional Manager	4 hours

Large	500–1,999 units	GHS 50,000–199,999	HQ Logistics Officer	8 hours
Strategic	≥ 2,000 units	≥ GHS 200,000	Director-General or Deputy	24 hours

The approval matrix is configurable by HQ Admin. All changes logged in audit trail.

12.3.3 Escalation Rules (SLA Breach)

Approver	SLA	Missed By	Escalates To
Regional Manager	4 hours	1 hour	HQ Logistics Officer
HQ Logistics Officer	8 hours	2 hours	Director-General
Director-General	24 hours	4 hours	Manual intervention flagged; no auto-approve

12.3.4 Digital Waybill Specification

Each approved transfer generates a digital waybill containing:

- Unique waybill number (e.g., `WB-ACC-SUN-2026-0041`)
- Transfer order reference
- Source and destination warehouse details
- Itemised list with SKU codes, descriptions, quantities, batch/lot numbers
- QR code encoding transfer ID + verification hash
- Vehicle registration, driver name, driver phone
- Expected delivery date and SLA
- Digital signature blocks for dispatcher and receiver
- PDF export (A4, print-ready)

12.3.5 Receipt Confirmation

1. Receiving officer scans waybill QR code or enters waybill number.
2. System displays dispatched quantities.
3. Officer counts physical items and enters received quantities.
4. If discrepancy detected, system prompts for photos and reason.
5. Officer captures digital signature (touch/stylus/mouse).
6. System updates stock, transfer status, and audit log.
7. Notifications sent to source warehouse, regional manager, and HQ logistics.

12.4 Module 6: Alerts & Notifications

12.4.1 Alert Categories

Category	Trigger	Default Recipients	Channels
Critical Stock	Stock ≤ minimum threshold	District Officer, Regional Manager, HQ Logistics	In-app + SMS + Email
Amber Stock	Stock ≤ 150% of minimum	District Officer, Regional Manager	In-app + SMS
Overdue Shipment	In-transit > SLA	Regional Manager, HQ Logistics, Source Officer	In-app + SMS + Email
Discrepancy	Quantity mismatch on receipt	Regional Manager, HQ Logistics, Auditor	In-app + SMS + Email
Approval SLA Breach	Approval pending > SLA	Next approver, HQ Admin	In-app + SMS + Email

Expiry Warning	Item expiring within 30/60/90 days	District Officer, Regional Manager	In-app + Email
System Security	Failed login, password reset, role change	System Admin, User	Email

12.4.2 Threshold Architecture

Thresholds operate at three levels, with the most specific level taking precedence:

Level	Who Sets It	Scope
System Default	Platform administrator	Applied to all warehouses for a given SKU category
Regional Override	Regional Manager	Overrides default for all districts in their region
District Override	District Officer (with Regional approval)	Overrides for a specific warehouse

12.4.3 Pre-loaded Recommended Defaults

Tier multipliers: District = 1x, Regional = 5x, HQ/Greater Accra = 20x

Category	Minimum (District)	Minimum (Regional)	Minimum (HQ)	Alert Logic
Food Packs	200 units	1,000 units	4,000 units	Red < min; Amber < 150% min
Shelter (Tents, Tarpaulins)	50 units	250 units	1,000 units	Red < min; Amber < 150% min
Medical Kits	30 units	150 units	600 units	Red < min; Amber < 150% min
Rescue Equipment	10 units	50 units	200 units	Red < min; Amber < 150% min
PPE	20 units	100 units	400 units	Red < min; Amber < 150% min
Water & Sanitation	100 units	500 units	2,000 units	Red < min; Amber < 150% min

Thresholds stored in `warehouse_thresholds` table: `warehouse_id`, `sku_category_id`, `min_quantity`, `amber_multiplier` (default 1.5), `set_by_user_id`, `set_at`. Most specific non-null row wins.

12.5 Module 7: Audit Trail & Supervision

12.5.1 Audit Log Requirements

Every audit log entry must capture:

- `event_id` — UUID
- `timestamp` — ISO 8601 UTC
- `user_id` — actor
- `user_role` — role at time of action
- `warehouse_id` — affected warehouse (if applicable)
- `region_id` — affected region (if applicable)
- `action` — CREATE / UPDATE / DELETE / LOGIN / LOGOUT / EXPORT / APPROVE / REJECT

- `entity_type` — transfer / stock / user / threshold / report / waybill
- `entity_id` — affected record ID
- `old_value` — JSON snapshot (for updates/deletes)
- `new_value` — JSON snapshot
- `ip_address` — client IP
- `user_agent` — browser/device
- `session_id` — session correlation
- `hash` — cryptographic hash chaining for tamper evidence

12.5.2 Tamper Evidence

- Audit log table is append-only at the database level (no UPDATE or DELETE).
- Each log entry includes a hash of the previous entry's hash + current payload.
- Daily hash snapshot exported to a separate write-once storage location.
- Any tampering attempt invalidates the hash chain and triggers a security alert.

12.5.3 Supervisory Drill-Down

HQ and authorised regional users can drill down:

```

National Overview
├── Region (e.g., Ashanti)
│   ├── District (e.g., Kumasi Metro)
│   │   ├── Warehouse
│   │   │   ├── Stock by SKU
│   │   │   ├── Transfer History
│   │   │   ├── Users
│   │   │   └── Audit Log

```

12.6 Module 10: Predictive Analytics & Intelligence

12.6.1 Demand Forecasting

The analytics engine will use:

- Historical consumption by region/district/SKU
- Seasonal disaster risk data (rainy season, dry season, Harmattan)
- Population density and vulnerability indices
- Current stock levels and expiry profiles
- Active and forecasted weather alerts from Ghana Meteorological Agency

Outputs:

- 30/60/90-day stock level projections
- Recommended pre-positioning quantities by region
- Risk-ranked warehouse list
- Automated procurement trigger suggestions

12.6.2 Stock Redistribution Optimiser

When a region faces a shortfall, the system recommends optimal source warehouses considering:

- Available surplus stock
- Distance / road conditions
- Vehicle availability
- expiry dates (FEFO)
- Historical transfer success rates

12.6.3 Expiry Risk Analysis

- Dashboard showing items expiring within 30/60/90 days.

- Recommendation to redistribute near-expiry stock to high-consumption districts.
- Alerts to warehouse officers to prioritise FEFO dispatch.

13. Data Model

13.1 Entity Relationship Overview

```
regions ||--o{ districts : contains
districts ||--o{ warehouses : contains
warehouses ||--o{ inventory : holds
warehouses ||--o{ warehouse_thresholds : configures
users ||--o{ user_warehouses : assigned_to
users ||--o{ transfers : creates
users ||--o{ transfers : approves
users ||--o{ transfers : receives
users ||--o{ audit_logs : generates
roles ||--o{ users : assigns
sku_categories ||--o{ skus : categorises
skus ||--o{ inventory : instances
skus ||--o{ transfer_items : moved_in
transfers ||--o{ transfer_items : contains
transfers ||--o{ waybills : generates
transfers ||--o{ gps_tracking : tracks
transfers ||--o{ discrepancies : raises
notifications ||--o{ users : sent_to
```

13.2 Core Tables

regions

Field	Type	Description
id	UUID PK	Unique region ID
code	VARCHAR(10)	ISO/Ghana region code (e.g., GR, AH, NR)
name	VARCHAR(100)	Region name (e.g., Greater Accra, Ashanti)
capital	VARCHAR(100)	Regional capital
geo_boundary	GEOJSON	Regional boundary polygon
risk_profile	JSONB	Seasonal risk scores by disaster type
created_at	TIMESTAMP	Record creation

districts

Field	Type	Description
id	UUID PK	Unique district ID
region_id	UUID FK	Parent region
name	VARCHAR(100)	District name
capital	VARCHAR(100)	District capital
geo_boundary	GEOJSON	District boundary polygon

population	INTEGER	Estimated population
vulnerability_index	DECIMAL	Composite vulnerability score

warehouses

Field	Type	Description
id	UUID PK	Unique warehouse ID
district_id	UUID FK	Parent district
code	VARCHAR(20)	Unique warehouse code (e.g., WH-GR-ACC)
name	VARCHAR(150)	Warehouse name
type	ENUM	hq / regional / district
address	TEXT	Physical address
latitude	DECIMAL	GPS latitude
longitude	DECIMAL	GPS longitude
manager_id	UUID FK	Assigned manager
capacity_m3	DECIMAL	Storage capacity in cubic metres
status	ENUM	operational / limited / closed
phone	VARCHAR(20)	Contact phone
email	VARCHAR(100)	Contact email

sku_categories

Field	Type	Description
id	UUID PK	Unique category ID
name	VARCHAR(50)	Category name
code	VARCHAR(20)	Category code
description	TEXT	Description
default_unit	VARCHAR(20)	Default unit of measure

skus

Field	Type	Description
id	UUID PK	Unique SKU ID
sku_code	VARCHAR(50) UK	Unique SKU code
name	VARCHAR(150)	Item name
category_id	UUID FK	Category
description	TEXT	Description
unit_of_measure	VARCHAR(20)	Unit (e.g., units, boxes, pallets)

weight_kg	DECIMAL	Weight per unit
volume_m3	DECIMAL	Volume per unit
shelf_life_days	INTEGER	Shelf life in days (optional)
hazard_class	VARCHAR(50)	Hazard classification (optional)
image_url	VARCHAR(255)	Item photo URL
is_active	BOOLEAN	Active status

inventory

Field	Type	Description
id	UUID PK	Inventory line ID
warehouse_id	UUID FK	Warehouse
sku_id	UUID FK	SKU
batch_lot	VARCHAR(50)	Batch or lot number
expiry_date	DATE	Expiry date (optional)
quantity	INTEGER	Current quantity
reserved_quantity	INTEGER	Quantity reserved for pending transfers
available_quantity	INTEGER	Computed: quantity - reserved
storage_location	VARCHAR(50)	Bin/zone location
last_counted_at	TIMESTAMP	Last physical count
updated_at	TIMESTAMP	Last update

warehouse_thresholds

Field	Type	Description
id	UUID PK	Threshold ID
warehouse_id	UUID FK	Warehouse (NULL for default)
region_id	UUID FK	Region (NULL unless regional override)
sku_category_id	UUID FK	Category
min_quantity	INTEGER	Minimum threshold
amber_multiplier	DECIMAL	Default 1.5
set_by_user_id	UUID FK	User who set it
set_at	TIMESTAMP	When set

transfers

Field	Type	Description
id	UUID PK	Transfer ID

transfer_number	VARCHAR(30) UK	Human-readable number (e.g., TRF-2026-0041)
source_warehouse_id	UUID FK	Source warehouse
destination_warehouse_id	UUID FK	Destination warehouse
created_by	UUID FK	Creator
created_at	TIMESTAMP	Creation time
status	ENUM	Transfer status
priority	ENUM	routine / urgent / emergency
approved_by	UUID FK	Approver
approved_at	TIMESTAMP	Approval time
vehicle_registration	VARCHAR(20)	Vehicle plate
driver_name	VARCHAR(100)	Driver name
driver_phone	VARCHAR(20)	Driver phone
dispatcher_id	UUID FK	Dispatcher
dispatched_at	TIMESTAMP	Dispatch time
received_by	UUID FK	Receiver
received_at	TIMESTAMP	Receipt time
expected_delivery_at	TIMESTAMP	Expected delivery
actual_delivery_at	TIMESTAMP	Actual delivery
digital_signature	TEXT	Base64 signature image
notes	TEXT	Notes

transfer_items

Field	Type	Description
id	UUID PK	Transfer item ID
transfer_id	UUID FK	Parent transfer
sku_id	UUID FK	SKU
batch_lot	VARCHAR(50)	Batch/lot
quantity_dispatched	INTEGER	Dispatched quantity
quantity_received	INTEGER	Received quantity
condition	ENUM	good / damaged / expired / missing

audit_logs

(See Module 7 for full field specification.)

13.3 Row-Level Security (RLS) Policies

Policy	Rule
District officers	SELECT/INSERT/UPDATE only where warehouse_id = assigned warehouse
Regional managers	Access where region_id = assigned region
HQ logistics	Full SELECT; INSERT/UPDATE on transfers, thresholds, warehouses
Director-General	Full SELECT; INSERT/UPDATE on approvals and executive actions
Auditor	Full SELECT only; no modifications
System admin	Full access for platform management
Audit log	INSERT only — no UPDATE or DELETE permitted by any role

14. API Design

14.1 API Architecture

- **Primary API:** Supabase PostgREST (auto-generated RESTful API from PostgreSQL schema)
- **Custom API:** Supabase Edge Functions for complex business logic (SMS, PDF generation, ML predictions, GPS)
- **Real-time:** Supabase Realtime for live dashboard updates
- **Authentication:** JWT tokens via Supabase Auth
- **Rate Limiting:** 100 requests/minute per user; 10 SMS/hour per user

14.2 Key Custom Endpoints

Endpoint	Method	Description	Auth
/functions/v1/transfer/create	POST	Create transfer order with validation	Authenticated
/functions/v1/transfer/approve	POST	Approve/reject transfer	Authenticated + authorised
/functions/v1/transfer/receive	POST	Confirm receipt with signature	Authenticated + authorised
/functions/v1/transfer/waybill	GET	Generate PDF waybill	Authenticated
/functions/v1/stock/intake	POST	Receive stock with audit	Authenticated
/functions/v1/stock/dispatch	POST	Dispatch stock	Authenticated
/functions/v1/stock/adjust	POST	Stock adjustment with approval	Authenticated + authorised
/functions/v1/alerts/check	POST	Trigger threshold alert evaluation	Service role
/functions/v1/sms/send	POST	Send SMS via Arkesel	Service role
/functions/v1/reports/generate	POST	Generate PDF/Excel report	Authenticated + authorised
/functions/v1/analytics/forecast	GET	Demand forecast	Authenticated + HQ/Regional
/functions/v1/gps/track	POST	Update vehicle GPS position	Authenticated + driver/field role
/functions/v1/import/bulk	POST	Bulk import users/SKUs/warehouses	Authenticated + admin

14.3 Real-Time Subscriptions

Channel	Event	Subscribers
public:transfers	INSERT/UPDATE/DELETE	Dashboard users, regional managers
public:inventory	UPDATE	Warehouse officers, dashboard
public:alerts	INSERT	Notification service, dashboard alert panel
public:gps_tracking	INSERT	Map viewers, logistics officers

15. Security Architecture

15.1 Defence-in-Depth Model

Layer 1: Perimeter Security TLS 1.3, WAF, DDoS protection, CDN
Layer 2: Application Security AuthN/AuthZ, MFA, RBAC, input validation
Layer 3: Data Security Encryption at rest (AES-256), RLS, field-level encryption for PII/sigs
Layer 4: Network Security VPC, private DB, IP allowlisting
Layer 5: Audit & Monitoring Immutable logs, SIEM, anomaly alerts
Layer 6: Operational Security Secrets management, least privilege, regular penetration testing

15.2 Security Requirements

Requirement	Implementation
Authentication	Username/password + TOTP MFA for privileged roles
Transport security	TLS 1.3 minimum; HSTS enabled; secure cookies
Data at rest	AES-256 encryption for database and object storage
Data in transit	TLS 1.3 for all API and database connections
Secrets management	HashiCorp Vault / AWS Secrets Manager / equivalent
Role-based access	Least privilege; enforced at application and database level
Row-level security	Supabase RLS policies prevent cross-tenant data leakage
Audit logging	Immutable, append-only, hash-chained audit log
Session security	Short-lived JWTs, refresh token rotation, inactivity timeout

Password policy	Min 12 chars, complexity, expiry, history restriction
API security	Rate limiting, input validation, CORS, CSRF protection
File uploads	Virus scanning, type validation, size limits, access controls
Backup security	Encrypted backups, off-site retention, access logging
Penetration testing	Annual third-party penetration test and vulnerability assessment

15.3 Data Classification

Classification	Examples	Handling
Public	Warehouse names, regional boundaries	No restriction
Internal	Stock levels, transfer schedules	Authenticated access
Confidential	User PII, driver details, donor information	Encrypted at rest, role-restricted
Restricted	Audit logs, security events, cryptographic keys	Admin-only, write-once storage

15.4 Incident Response

Severity	Examples	Response Time
Critical	Data breach, unauthorised admin access, system-wide outage	1 hour
High	Suspected account compromise, mass failed logins	4 hours
Medium	Policy violation, non-critical vulnerability	24 hours
Low	Minor access request, documentation update	72 hours

16. Compliance & Governance

16.1 Regulatory Compliance

Regulation / Standard	Requirement	NADMO-WMS Compliance
Ghana Data Protection Act, 2012 (Act 843)	Lawful processing, data minimisation, security, individual rights	Privacy-by-design; data minimisation; encryption; access controls; audit trail
NITA ICT Standards	Government IT standards, interoperability, security	Standards-based APIs; security architecture review; NITA certification path
Public Financial Management Regulations	Accountability, audit trail, asset tracking	Immutable transaction logs, valuation reports, reconciliation tools
GIFMIS Integration (Future)	Financial reporting alignment	Standard chart of accounts mapping; export formats
UN OCHA Humanitarian Data Principles	Transparency, privacy, ethics	Donor reporting, anonymised analytics, ethical use guidelines

16.2 Data Residency & Sovereignty

- Phase 1: Data hosted in Supabase Cloud (region selectable; preference for EU or African region with migration path).

- Phase 2: Containerised deployment option for GovTech Ghana or accredited local cloud.
- Phase 3: Full sovereign hosting in Ghana with NITA accreditation.
- All data remains property of the Government of Ghana at all times.
- Full data export available on demand in open formats (CSV, JSON, PDF).

16.3 Accessibility

- WCAG 2.1 AA compliance target.
- Keyboard-navigable interface.
- Screen reader compatible components.
- Colour contrast ratios meet accessibility standards.
- Mobile-responsive design for low-resolution screens.

16.4 Ethical Use & Bias

- Predictive analytics recommendations are advisory; human approval required for all dispatches.
- Algorithmic decisions are explainable and auditable.
- No automated decision-making affecting beneficiary rights.
- Regular review of model fairness across regions and disaster types.

17. Design System & User Experience

17.1 Design Principles

Principle	Application
Clarity under pressure	Large KPIs, clear status colours, minimal cognitive load for crisis use
Mobile-first field operations	Touch-friendly buttons, offline awareness, simple workflows
Trust through transparency	Every action logged, every status visible, every alert traceable
Ghana-centric	Local terminology, Ghana map, regional context, multi-language ready
Professional government grade	Clean, authoritative, no unnecessary decoration

17.2 Colour Palette

Token	Hex	Usage
--nadmo-red	#C41E3A	Critical alerts, emergencies, destructive actions
--nadmo-amber	#F59E0B	Warnings, pending approvals, amber stock
--nadmo-green	#10B981	OK status, received, success
--nadmo-blue	#0066CC	Primary brand, in-transit, links
--nadmo-navy	#0F172A	Headers, text, sidebar
--nadmo-grey-100	#F1F5F9	Backgrounds
--nadmo-grey-500	#64748B	Secondary text
--nadmo-white	#FFFFFF	Cards, surfaces

17.3 Typography

- **Headings:** Inter or system sans-serif (clean, highly legible)
- **Body:** Inter / Roboto
- **Monospace:** JetBrains Mono for data tables and audit logs

- **Base size:** 16px; minimum touch target 44×44px

17.4 Iconography

- Use Lucide icons or equivalent for consistency.
- Status icons:
 -  Green check for OK/Received
 -  Amber triangle for Warning
 -  Red circle for Critical
 -  Blue truck for In-Transit
 -  Grey clock for Pending

17.5 Component Library

- Built on **shadcn/ui** + **Radix UI** primitives.
- Custom components: status badges, alert cards, transfer timeline, signature pad, map choropleth, KPI tiles, data tables with pagination.

17.6 User Experience Tenets

1. **Three-click rule:** Core actions (receive, dispatch, transfer) achievable within three clicks from the dashboard.
2. **Progressive disclosure:** Advanced options hidden until needed.
3. **Optimistic feedback:** Actions show immediate UI feedback; background sync for slower operations.
4. **Error forgiveness:** Clear error messages, undo where safe, autosave on forms.
5. **Contextual help:** Tooltips, inline guidance, and a searchable help centre.

18. Connectivity Resilience — Low-Bandwidth Design

Ghana's connectivity varies significantly between urban HQ offices and rural district warehouses. NADMO-WMS is designed to function on **1–2 Mbps connections as a baseline**, with full offline capabilities in Phase 2.

Principle	Implementation
Lazy loading	Heavy assets (maps, large tables) load after critical UI
Pagination	All lists paginated (default 25 rows/page; max 100)
Optimistic UI	Form submissions show immediate feedback; sync in background
Compressed assets	WebP images; JS/CSS minified + Brotli/gzip via CDN
Progressive data loading	Dashboard KPI tiles load independently; skeleton screens
Request caching	Read-heavy data cached locally for 5 minutes via service worker
No heavy media	Zero autoplay media; SVG charts preferred over canvas
Connection status	Banner when offline: "You are offline. Changes will sync when reconnected."
Background sync	PWA service worker queues mutations and syncs on reconnection
Data sync conflict resolution	Last-write-wins with manual merge for critical conflicts

19. Ghana National Map & GIS

19.1 Map Requirements

Item	Detail
------	--------

Map library	Mapbox GL JS (recommended) or Leaflet.js
Base tiles	OpenStreetMap / Mapbox Streets
Boundary data	OSM administrative boundaries — admin level 4 (regions), admin level 6 (districts) via Geofabrik Ghana extract
Choropleth colours	Green (all OK) / Amber (warning) / Red (critical) / Grey (no data)
Interactions	Click district → detail panel slides in; hover → tooltip (district name + critical item count)
Data refresh	Map colours refresh every 5 minutes or on real-time event
Low-connectivity fallback	If tiles fail, degrade to tabular district list sorted by criticality
Vehicle tracking	Live GPS positions overlaid on map for in-transit shipments
Attribution	"© OpenStreetMap contributors" required (ODbL licence)

19.2 GhanaPost GPS Integration

- Use GhanaPost GPS addresses for warehouse location capture and waybill delivery coordinates.
- Fallback to manual latitude/longitude entry for locations not yet covered.

20. Copy & Language Standards

Overall Tone: Professional, Authoritative, Calm under Pressure — Government emergency management platform. Clear, direct, institutional language throughout.

Context	Copy
Page title — Dashboard	"National Operations Overview"
Page title — Transfers	"Transfer Orders"
Empty state — No transfers	"No transfer orders yet. Create your first transfer to begin tracking logistics movement."
Empty state CTA	"Create Transfer Order"
Success — Transfer created	"Transfer Order #[ID] created successfully. Waybill is ready to download."
Error — Insufficient stock	"Insufficient stock at [Warehouse Name]. Available: [X] units. Requested: [Y] units."
Error — Network failure	"Connection lost. Your progress has been saved. Please retry when your connection is restored."
Loading state	"Loading operational data..."
Cancel transfer confirm	"Cancel this transfer? The destination warehouse will be notified. This cannot be undone."
Cancel confirm button	"Yes, Cancel Transfer"
Cancel abort button	"Keep Transfer Active"
Discard changes prompt	"You have unsaved changes. Discard them?"

Low stock alert	"[Warehouse Name] — [Item Name] is below the minimum threshold ([X] units remaining)."
Overdue shipment alert	"Transfer #[ID] to [Warehouse] is overdue. Expected delivery: [Date]. Please investigate."
Discrepancy raised	"Quantity discrepancy on Transfer #[ID]. Dispatched: [X]. Received: [Y]. Escalated to your Regional Manager."
Receive confirmation CTA	"Confirm Receipt & Sign"
Export button	"Export Report"
No permission error	"You do not have permission to perform this action. Contact your administrator."
Offline banner	"You are offline. Changes will sync automatically when your connection is restored."

20.1 Localisation Roadmap

- **Phase 1:** English (Ghana)
 - **Phase 2:** Twi, Ga, Ewe, Hausa, Dagbani for field-facing mobile screens
 - **Phase 3:** Full platform localisation
-

21. Non-Functional Requirements

21.1 Performance

Metric	Requirement
First Contentful Paint	< 2.0 seconds on 5 Mbps connection
Time to Interactive	< 4.0 seconds on desktop; < 6.0 seconds on mobile 3G
Dashboard data refresh	≤ 5 minutes (configurable); manual refresh always available
Real-time event propagation	< 3 seconds from transaction to dashboard update
Transfer order submission	API response < 800 ms
Report generation (national)	< 10 seconds for 90-day range; < 30 seconds for annual
Concurrent users	500+ simultaneous users without degradation
Peak load handling	2× normal concurrent load during national emergencies
Uptime SLA	99.9% excluding scheduled maintenance
Scheduled maintenance	Off-peak windows (02:00–05:00 GMT); 48-hour notice

21.2 Scalability

Dimension	Target
Warehouses	300+ (HQ + 16 regions + 261+ districts)
SKUs	10,000+ active items
Users	1,000+ active users
Transactions	100,000+ per month

Audit log	10 million+ records with query optimisation
Reports	National reports generated in < 10 seconds

21.3 Reliability

Requirement	Implementation
Database backups	Automated every 6 hours; daily full backup; 30-day rolling retention
Off-site backup	Encrypted copies to geographically separate storage
Disaster recovery	RPO ≤ 1 hour; RTO ≤ 4 hours
Health checks	Automated uptime and dependency monitoring
Circuit breakers	Graceful degradation when external services (SMS, GPS) fail

21.4 Security (Detailed)

Requirement	Implementation
Authentication	Username + Password + TOTP MFA
Transport security	TLS 1.3; HTTPS enforced; HSTS
Data at rest	AES-256 encryption (database + backups + object storage)
Data in transit	TLS 1.3 for all connections
Role-based access	Least privilege; enforced at application and DB level
Audit logging	Immutable, append-only, hash-chained log
Session security	30-minute inactivity timeout; single active session policy; token rotation
Password policy	Min 12 chars, complexity, 90-day expiry, no reuse of last 6
Hosting	Vercel + Supabase (Phase 1); GovTech Ghana self-hosted (Phase 2)
Vulnerability management	Monthly dependency scans; annual penetration test

21.5 Compatibility

Dimension	Requirement
Desktop browsers	Chrome 100+, Edge 100+, Firefox 100+, Safari 15+
Mobile browsers	Chrome on Android 9+, Safari on iOS 14+
Minimum screen	320px width (mobile-first responsive)
Tablets	Optimised for 7–10 inch Android tablets
Language	English (Phase 1); Twi, Ga, Ewe, Hausa, Dagbani (Phase 2)

21.6 Data & Retention

Item	Policy
Transaction records	Retained indefinitely (government compliance)

Deleted records	Soft-delete only — never purged
Audit logs	Retained indefinitely; write-once storage
Backups	Daily automated; 30-day rolling retention; off-site copy
Data ownership	All data owned by NADMO / Government of Ghana
Export	Full data export available at any time (CSV/JSON/PDF) — no vendor lock-in
Data purge	No automatic purge; archive-only after 7 years

22. Implementation Roadmap

22.1 Phase 1: Foundation & National Visibility (Months 1–6)

Month	Deliverables
1	Project kick-off, stakeholder alignment, detailed technical design, security architecture review
2	Core platform setup (Next.js + Supabase), authentication, RBAC, warehouse/SKU master data
3	Inventory management, transfer orders, waybill generation, digital signatures
4	National/regional/district dashboards, alerts, SMS integration (Arkesel), audit trail
5	Reporting module, bulk import, user management, connectivity resilience, UAT preparation
6	User acceptance testing, security audit, training rollout, pilot launch (Greater Accra + 3 regions)

22.2 Phase 2: Scale & Intelligence (Months 7–12)

Month	Deliverables
7	National rollout to all 16 regions and 261 districts
8	Native Android app with offline sync; hardware barcode scanner support
9	GPS vehicle tracking integration; GhanaPost GPS
10	Predictive analytics and demand forecasting
11	Advanced reports, donor templates, cold-chain monitoring (pilot)
12	GovTech Ghana migration readiness assessment; sovereign hosting pilot

22.3 Phase 3: Ecosystem Integration (Year 2+)

Initiative	Description
GIFMIS Integration	Financial data exchange and procurement alignment
NEMA / Met Agency Integration	Incident linking and weather-based risk alerts
Beneficiary Distribution	Track aid from warehouse to final beneficiary
IoT Cold Chain	Temperature/humidity sensors in strategic warehouses
Multi-Language	Full Twi, Ga, Ewe, Hausa, Dagbani support
Open API	Government-approved API for partner systems

22.4 Go-Live Strategy

Wave	Scope	Timeline
Pilot Wave 1	Greater Accra HQ + 3 regional warehouses + 12 district warehouses	Month 6
Wave 2	All 16 regional warehouses + 50 district warehouses	Month 8
Wave 3	All 261 district warehouses	Month 10
National Operations	Full operational use; decommission paper registers	Month 12

23. Success Metrics & Key Performance Indicators (KPIs)

23.1 Operational KPIs

KPI	Baseline	Target (12 Months)	Measurement
% warehouses digitised	0%	100%	Active warehouse accounts
Stock data accuracy	Unknown	≥ 98%	Cycle count variance
Average dispatch-to-receipt time	Days	≤ 24 hours (intra-region)	Transfer lifecycle data
% transfers with digital waybill	0%	100%	Transfer records
% receipts confirmed within SLA	0%	≥ 95%	Receipt timestamps
Critical stock response time	Hours/days	≤ 2 hours	Alert-to-action time

23.2 User Adoption KPIs

KPI	Target
Active users (monthly)	≥ 90% of registered users
Transactions captured digitally	≥ 95% of all stock movements
Mobile/tablet usage	≥ 70% of district/field users
Training completion	≥ 95% of target users
User satisfaction (CSAT)	≥ 4.0 / 5.0

23.3 Governance & Accountability KPIs

KPI	Target
Audit trail coverage	100% of create/update/delete actions
Discrepancy resolution time	≤ 48 hours
Report generation time	≤ 2 minutes for standard reports
Data export availability	100% on demand
Security incidents	Zero critical breaches

23.4 Impact KPIs

KPI	Baseline	Target
Response time to disaster-affected communities	Variable	50% reduction
Stockout incidents before disasters	Unknown	80% reduction
Relief item losses / unaccounted items	Unknown	90% reduction
Donor reporting preparation time	Weeks	≤ 2 days

24. Risk Register

#	Risk	Likelihood	Impact	Mitigation
1	Poor internet connectivity in rural districts	High	High	PWA offline mode; low-bandwidth design; SMS fallback; scheduled sync windows
2	Low user adoption / resistance to change	High	High	Change management; training; super-user network; executive sponsorship; simple UX
3	Data entry errors during digitisation	Medium	High	Bulk import validation; barcode/QR scanning; mandatory fields; cycle counts
4	Cybersecurity breach / unauthorised access	Low	Critical	MFA, RBAC, RLS, encryption, penetration testing, security monitoring
5	Vendor lock-in / hosting dependency	Medium	Medium	Cloud-agnostic architecture; open-source stack; data export; migration plan
6	Integration delays with external systems	Medium	Medium	Well-documented APIs; phased integration; fallback manual processes
7	Device availability in districts	Medium	High	PWA works on existing phones/tablets; procurement plan for district tablets
8	Power outages affecting field use	High	Medium	Offline-first mobile app; power banks; low-power devices
9	Inaccurate master data (warehouses, SKUs)	Medium	High	Data cleansing workshop; validation rules; regional review process
10	Budget / procurement delays	Medium	High	Phased implementation; clear MVP scope; strong business case

25. Quality Assurance & Testing Strategy

25.1 Testing Levels

Level	Scope	Tools
Unit Testing	Functions, utilities, business logic	Jest, Vitest
Integration Testing	API endpoints, database policies, auth flows	Playwright API, Postman
End-to-End Testing	Critical user journeys	Playwright
Accessibility Testing	WCAG 2.1 AA compliance	axe-core, Lighthouse

Performance Testing	Load, stress, low-bandwidth simulation	k6, Lighthouse
Security Testing	Vulnerability scans, penetration test	OWASP ZAP, Burp Suite, third-party pentest
User Acceptance Testing	NADMO stakeholders pilot real scenarios	Manual, scripted scenarios

25.2 Critical Test Scenarios

1. User login with MFA across all roles.
2. Stock intake and dispatch with barcode/QR scanning.
3. Transfer order creation, approval, dispatch, tracking, and receipt.
4. Discrepancy reporting and reconciliation.
5. Critical stock alert generation and SMS delivery.
6. National dashboard drill-down and report export.
7. Offline data capture and background sync.
8. RLS policy enforcement — users cannot access unauthorised warehouses.
9. Audit log immutability and hash chain integrity.
10. Bulk import of users, SKUs, and warehouses.

25.3 UAT Acceptance Criteria

- 100% of core user stories pass UAT.
- Zero critical or high-severity defects.
- ≤ 5 medium-severity defects with documented workarounds.
- User satisfaction score $\geq 4.0 / 5.0$.
- System performance meets NFRs under UAT load.

26. Training & Change Management

26.1 Training Programme

Audience	Format	Duration	Content
HQ Executives	Executive briefing	2 hours	Dashboards, reports, approvals
Regional Managers	Workshop + hands-on	1 day	Regional oversight, approvals, discrepancy resolution
District Officers	Field training	2 days	Stock intake/dispatch, transfers, mobile usage, offline sync
Field Officers	Field training	1 day	Mobile app, GPS, photo capture, signatures
Auditors	Workshop	1 day	Audit logs, reports, exports
System Admins	Technical training	3 days	User management, security, backups, integrations

26.2 Change Management Approach

- **Executive sponsorship:** Director-General champions the platform.
- **Super-user network:** At least one super-user per region and district.
- **Help desk:** Dedicated support channel during rollout.
- **Quick reference guides:** Laminated cards for district warehouses.
- **Feedback loop:** Monthly user feedback sessions and product improvements.
- **Incentives:** Recognition for high-adoption districts and accurate data.

26.3 Training Materials

- Video tutorials (English + local languages).
- Interactive user manual embedded in the app.
- Printable quick-start guides.
- Sandbox environment for practice.
- FAQ and troubleshooting knowledge base.

27. Support & Maintenance

27.1 Support Tiers

Tier	Issue Type	Response Time	Resolver
L1	Password resets, login issues, basic how-to	≤ 4 hours	NADMO IT / Helpdesk
L2	Feature bugs, data issues, report problems	≤ 8 hours	Vendor support team
L3	Security incidents, performance degradation, integration failures	≤ 2 hours	Vendor engineering + NADMO IT

27.2 Maintenance Windows

- Routine updates: Monthly, scheduled off-peak.
- Security patches: As needed, with emergency change process.
- Major releases: Quarterly, with 2-week notice.

28. Budget & Estimation Framework

Note: The following is an indicative framework for procurement planning. Detailed costs will be provided in the commercial proposal.

28.1 Cost Categories

Category	Description
Software Development	Design, frontend, backend, mobile app, integrations
Cloud Infrastructure	Hosting, database, storage, CDN, SMS, monitoring (Phase 1)
Security & Compliance	Penetration testing, security audit, compliance review
Data Migration & Onboarding	Master data cleansing, bulk import, training
Change Management	Training materials, workshops, super-user network
Support & Maintenance	L2/L3 support, updates, enhancements
Hardware (Optional)	Tablets, barcode scanners, GPS trackers, IoT sensors

28.2 indicative Team Structure

Role	Effort
Product Manager	Full-time
Solutions Architect	Full-time (Months 1–3), part-time thereafter

UX/UI Designer	Full-time (Months 1–4)
Frontend Engineers (Next.js)	2 full-time
Backend Engineers (Supabase/PostgreSQL)	2 full-time
Mobile Engineer (Android/Flutter)	1 full-time (Months 4–10)
DevOps / Security Engineer	1 full-time
QA Engineer	1 full-time
Data / ML Engineer	Part-time (Months 6–12)

29. SMS Notification Templates (Arkesel)

Provider: Arkesel (arkesel.com) — Ghana-local SMS API

Sender ID: NADMO-WMS (registered branded sender ID)

Event	Recipients	Message
Critical stock alert	District Officer, Regional Manager, HQ Logistics	NADMO ALERT: [Warehouse] critically low on [Item]. [X] units left. Act: wms.nadmo.gov.gh
Amber stock warning	District Officer, Regional Manager	NADMO WARNING: [Warehouse] [Item] approaching min level ([X] units). Review stock.
Transfer approved	Requestor	Transfer #[ID] APPROVED by [Name]. Assign vehicle now. NADMO-WMS.
Transfer rejected	Requestor	Transfer #[ID] REJECTED by [Name]. Reason: [reason]. Login for details.
Transfer dispatched	Destination Officer, Regional Manager	Transfer #[ID] dispatched to [Warehouse]. Expected: [Date]. Waybill ready.
Transfer overdue	Regional Manager, HQ Logistics	OVERDUE: Transfer #[ID] to [Warehouse]. Expected: [Date]. Investigate now.
Discrepancy raised	Regional Manager, HQ Logistics, Auditor	DISCREPANCY: Transfer #[ID]. Dispatched [X], received [Y]. Action required.
Approval SLA breach	Next approver, HQ Admin	ACTION REQUIRED: Transfer #[ID] awaiting approval (escalated). wms.nadmo.gov.gh
Receipt confirmed	Source Officer, HQ Logistics	Transfer #[ID] received at [Warehouse]. [X] items confirmed. NADMO-WMS.
Expiry warning	District Officer, Regional Manager	NADMO: [Item] at [Warehouse] expires in [X] days. Prioritise dispatch.

Technical Notes:

- API: Arkesel SMS API v2 — `POST /api/v2/sms/send`
- Messages kept $\leq$ 160 characters where possible (single SMS unit).
- Phone numbers normalised to E.164 format (`+233XXXXXXXX`).
- Rate limit: max 10 SMS per user per hour; max 100 SMS per organisation per hour.
- Fallback: if Arkesel unreachable, event queued and retried every 5 minutes; in-app notification always fires.

30. Bulk Data Import Specifications

30.1 Import Types Available from Launch

Data Type	Format	Template Fields
SKU Catalogue	CSV/Excel	sku_code, name, category_code, unit_of_measure, description, weight_kg, volume_m3, shelf_life_days, image_url
Warehouse Directory	CSV/Excel	code, name, type, region_code, district_name, address, latitude, longitude, manager_email, phone, capacity_m3
User Accounts	CSV/Excel	email, phone, first_name, last_name, role_code, warehouse_codes, region_code
Opening Stock Balances	CSV/Excel	warehouse_code, sku_code, batch_lot, quantity, expiry_date, storage_location

30.2 Import Validation Rules

- Duplicate codes flagged before import.
- Missing required fields highlighted.
- Invalid references (e.g., non-existent warehouse) rejected.
- Preview mode showing valid (green), warning (amber), error (red) rows.
- Partial import: valid rows imported; error rows skipped with downloadable error report.
- Max file size: 10 MB; max rows: 10,000 per import.
- All imports logged in audit trail.

31. Open Items & Decisions Log

#	Item	Decision	Owner	Status
1	Threshold defaults	System-configurable with pre-loaded defaults per tier; editable by authorised users	HQ Logistics	Resolved
2	Approval workflow	Hierarchical 4-tier matrix (Routine → Regional → HQ → Director); editable by HQ Admin	DG Office	Resolved
3	SMS gateway	Arkesel (Ghana-local)	IT	Resolved
4	Hosting environment	Vercel + Supabase Cloud (Phase 1); GovTech Ghana migration path (Phase 2)	NITA / NADMO	Resolved
5	SKU catalogue	Manual entry + CSV/Excel bulk import from launch	HQ Logistics	Resolved
6	Existing data migration	No legacy data at launch; bulk import capability available at any time	Project Team	Resolved
7	Internet connectivity	Low-bandwidth resilience; full offline sync in Phase 2	Architecture	Resolved
8	Ghana mapping data	OpenStreetMap (OSM) with Mapbox/Leaflet	Architecture	Resolved
9	GPS vehicle tracking	Core Phase 1 feature via driver mobile app + GhanaPost GPS	Architecture	Resolved
10	Barcode/QR scanning	Core Phase 1 via PWA camera + optional Bluetooth scanner	Architecture	Resolved

11	MFA enforcement	Required for HQ, Regional, Auditor; recommended for District	Security	Resolved
12	Predictive analytics	Phase 2 ML-driven forecasting and recommendations	Data/ML	Resolved

32. Appendices

Appendix A: Glossary

Term	Definition
FEFO	First-Expiry-First-Out — stock dispatch method prioritising items closest to expiry
GIFMIS	Ghana Integrated Financial Management Information System
MFA	Multi-Factor Authentication
MMDA	Metropolitan, Municipal, and District Assembly
NADMO	National Disaster Management Organisation
NEMA	National Emergency Management Organisation (Ghana)
NITA	National Information Technology Agency
PWA	Progressive Web Application
RLS	Row-Level Security (database access control)
SKU	Stock Keeping Unit
SLA	Service Level Agreement
TOTP	Time-based One-Time Password
WMS	Warehouse Management System

Appendix B: Reference URLs

- NADMO: <https://www.nadmo.gov.gh>
- NITA: <https://nita.gov.gh>
- Ghana Data Protection Commission: <https://dataprotection.gov.gh>
- Ghana Meteorological Agency: <https://www.meteo.gov.gh>
- GhanaPost GPS: <https://www.ghanapostgps.com>
- OpenStreetMap: <https://www.openstreetmap.org>
- Arkesel: <https://arkesel.com>
- Supabase: <https://supabase.com>

Appendix C: Document History

Version	Date	Author	Changes
1.0	May 2026	Initial Team	First draft
1.1	June 2026	Product Team	Resolved open items, added hosting details

2.0	June 2026	Solutions Architecture Team	Complete rewrite: architecture, security, AI/ML, compliance, roadmap, KPIs, risk register
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33. Conclusion

The NADMO Integrated Warehouse & Logistics Management System (NADMO-WMS) represents a transformational investment in Ghana’s disaster preparedness and response capability. By digitising every warehouse, tracking every movement, and providing real-time national visibility, NADMO-WMS will:

- Save lives by ensuring faster, more coordinated disaster response.
- Protect public resources through immutable accountability and reduced losses.
- Strengthen donor confidence with transparent, instant reporting.
- Position NADMO as a regional leader in humanitarian logistics technology.

This PRD provides a comprehensive, procurement-ready blueprint for building an award-winning, government-grade logistics platform. The architecture is secure, scalable, sovereign, and designed for the realities of Ghana’s operating environment — from high-speed HQ offices to low-bandwidth district warehouses.

Next Steps:

1. Executive review and approval of PRD v2.0.
2. Technical architecture deep-dive with NITA and NADMO IT.
3. Commercial proposal and procurement process.
4. Project kick-off and Phase 1 development.

NADMO-WMS PRD v2.0 — Ready for Development, Procurement, and Government Review.

Prepared for: National Disaster Management Organisation (NADMO), Republic of Ghana

Classification: Confidential — For Government Review